

Power BI Assignment 2 – DAX & Data Visualization

In this assignment DAX (Data Analysis Expressions) is used to analyse and transform E-commerce sales data, generating key metrics and insights. Hereby carefully analysed the dataset ensuring the visualisation gain valuable insights into sales trends, profitability, and performance metrics within the e-commerce domain.

About Dataset

Tables	Rows	Columns
List of orders	500	6
Order Details	500	11
Order Date	1,33,500	4
Sale target	36	4

Let's see the Screenshot of tasks for data modelling and data visualisation

Calculated column

For creating calculated column's the DAX function's I used are pointed below

- Sales_Category = IF('Order Details'[Amount]>500,"Above Average","Below Average")
- Revenue Per Order = 'Order Details'[Amount]*'Order Details'[Quantity]
- Category_type = 'Order Details'[Category] & " -" & 'Order Details'[Sub-Category]

The screenshot shows the DAX Studio interface with a calculated column formula: `Category_type = 'Order Details'[Category] & " -" & 'Order Details'[Sub-Category]`. The table below represents the data shown in the screenshot.

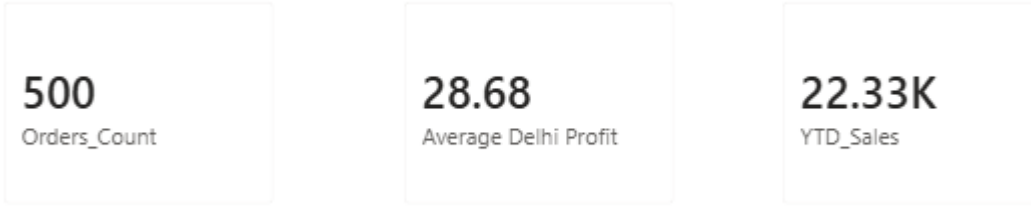
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category_type	Revenue Per Order	Sales_Category
5656	6	3	1	Clothing	Hankerchief	0.5	Profit	Clothing -Hankerchief	6	Below Average
5873	7	1	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing -Hankerchief	7	Below Average
5851	10	2	2	Clothing	Hankerchief	0.2	Profit	Clothing -Hankerchief	20	Below Average
6088	11	5	2	Clothing	Hankerchief	0.454545454545454	Profit	Clothing -Hankerchief	22	Below Average
5979	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing -Hankerchief	24	Below Average
5603	12	1	2	Clothing	Hankerchief	0.0833333333333333	Profit	Clothing -Hankerchief	24	Below Average
5940	13	3	2	Clothing	Hankerchief	0.230769230769231	Profit	Clothing -Hankerchief	26	Below Average
5991	13	5	2	Clothing	Hankerchief	0.384615384615385	Profit	Clothing -Hankerchief	26	Below Average
5670	14	2	1	Clothing	Hankerchief	0.142857142857143	Profit	Clothing -Hankerchief	14	Below Average
5757	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit	Clothing -Hankerchief	14	Below Average
6097	14	5	1	Clothing	Hankerchief	0.357142857142857	Profit	Clothing -Hankerchief	14	Below Average
6070	14	7	2	Clothing	Hankerchief	0.5	Profit	Clothing -Hankerchief	28	Below Average
5784	15	4	1	Clothing	Hankerchief	0.266666666666667	Profit	Clothing -Hankerchief	15	Below Average
6056	15	6	2	Clothing	Hankerchief	0.4	Profit	Clothing -Hankerchief	30	Below Average
5997	16	6	3	Clothing	Hankerchief	0.375	Profit	Clothing -Hankerchief	48	Below Average
5848	16	8	2	Clothing	Hankerchief	0.5	Profit	Clothing -Hankerchief	32	Below Average
5864	17	5	1	Clothing	Hankerchief	0.294117647058824	Profit	Clothing -Hankerchief	17	Below Average
5965	17	7	3	Clothing	Hankerchief	0.411764705882353	Profit	Clothing -Hankerchief	51	Below Average
5793	18	1	3	Clothing	Hankerchief	0.0555555555555556	Profit	Clothing -Hankerchief	54	Below Average
6010	18	2	3	Clothing	Hankerchief	0.111111111111111	Profit	Clothing -Hankerchief	54	Below Average
5888	18	2	3	Clothing	Hankerchief	0.111111111111111	Profit	Clothing -Hankerchief	54	Below Average
6079	18	3	2	Clothing	Hankerchief	0.166666666666667	Profit	Clothing -Hankerchief	36	Below Average

Calculated Measures

The DAX function's used in calculated are pointed below

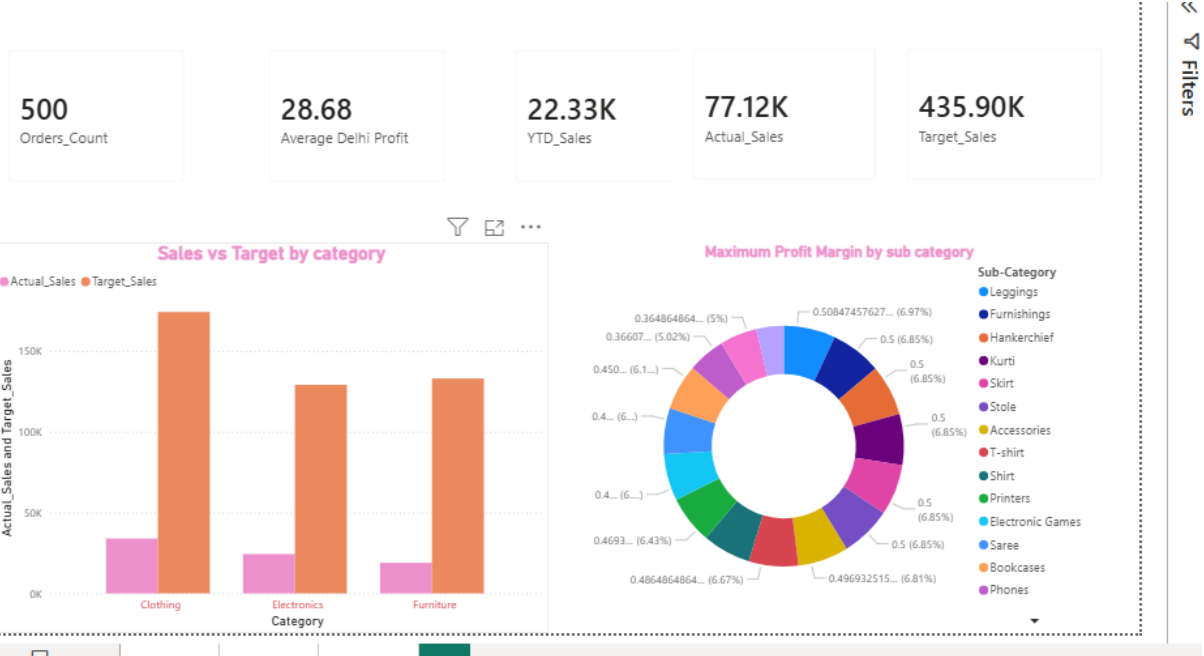
- Orders_Count = COUNT('Order Details'[Order ID])
- Average Delhi Profit = CALCULATE(AVERAGE('Order Details'[Profit]),'List of Orders'[State]="Delhi")

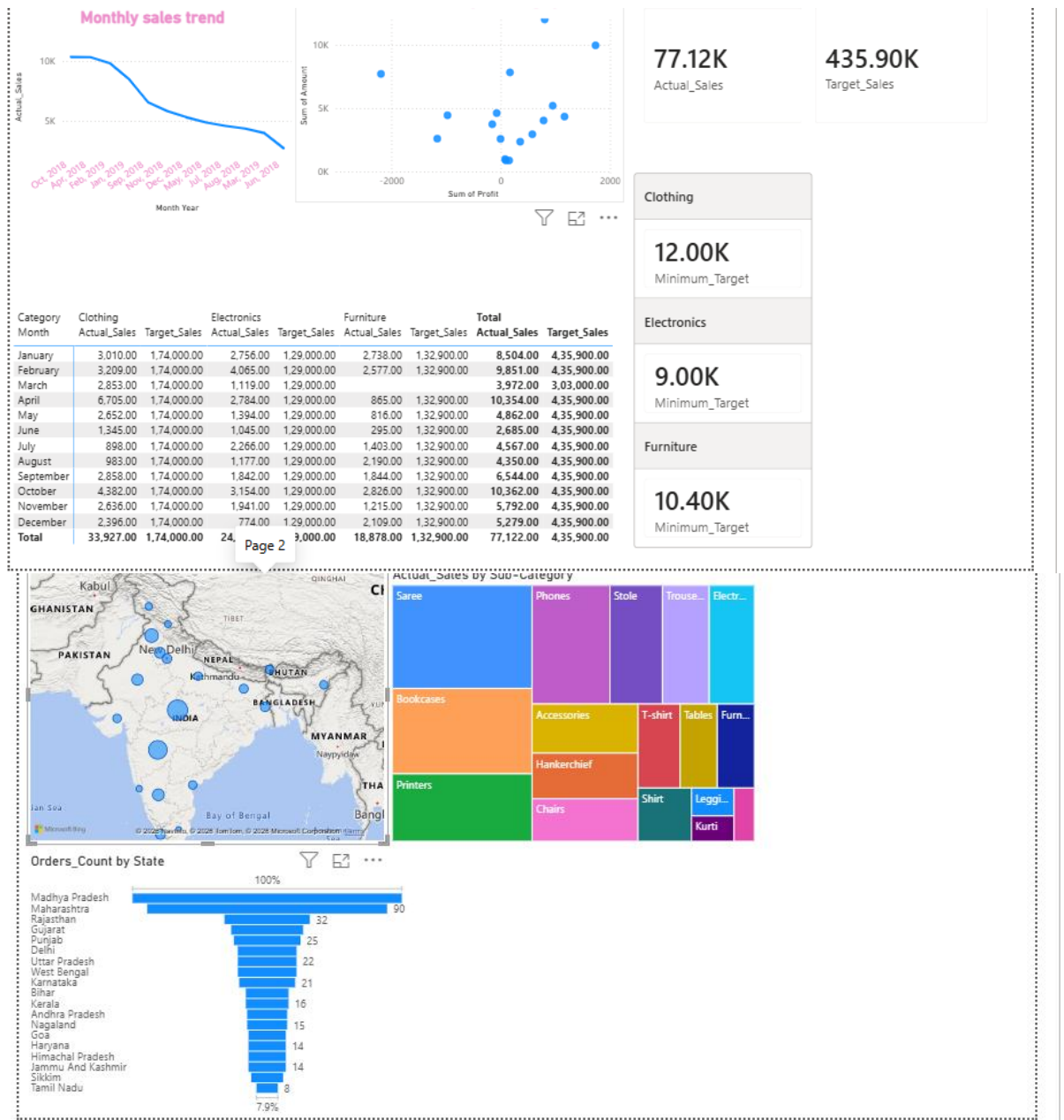
➤ $\text{YTD_Sales} = \text{CALCULATE}(\text{SUM}('Order\ Details'[Amount]), \text{DATESYTD}('List\ of\ Orders'[Order\ Date].[Date]))$



Data Visualisation

Using E-Commerce dataset, various visualisation were done using multiples of chart. Profit analysis, monthly sales trends, Target and sales comparisons, top sales based on city, Sale performances matrices etc were done to create meaningful insights.





Conclusion

After a detailed analysis and visualisation, this organisation has not achieved target sale in all 3 category. Also profit margin is low in most of the product sale.

- From cluster column chart, what we observe that is actual sales is lower than target sales.
- From Matrix table we can observe that sales of clothing better when compares to rest of two category
- From tree chart, we can say that saree is the most sold sub- category product
- From tunnel chart, Madhya Pradhesh, Bhopal has the order count compared to others
- From Scatter chart, we can mention that among all other months Oct, 2018 has the highest sale
- From donut chart, we can say legging has the highest profit margin

